

MVP-01 — Objective Crystallizer

PRODUCT.md

WEDGE · Learning OS — Strategic Translator Step 01

P1 — Founding Premise

The behavioral fact: Teachers open sessions with intent. Intent is not an objective. "I want them to understand conditionals" cannot be falsified — there is no state of the world at session end that constitutes pass or fail. Without a falsification criterion, the teacher cannot know whether the session worked, cannot isolate what to optimize, and cannot replicate the conditions that produced learning.

The mechanism gap: Existing approaches (lesson plan templates, curriculum frameworks, learning outcome banks) describe content to deliver. None force the teacher to declare what observable, countable behavior constitutes proof of understanding. They produce well-structured sessions with no north star.

The enabling insight: A falsifiable objective is not a more specific version of an intent. It is a structurally different object: it names what to maximize, what must not be compromised in pursuit of that, and what observable criterion constitutes proof. Teachers can state intent fluently. They cannot derive this structure under time pressure.

The distinctive mechanism: Teacher states intent in any form. System returns one constraint-form objective — three fields only: maximize / must not break / success criterion. One API call. No iteration.

Acceptable Sacrifice: Teachers whose pedagogy is relationship- or atmosphere-driven and who treat explicit outcome measurement as reductive. Teachers who run exploratory or discussion-based sessions where behavioral success criteria are inappropriate. This tool does not serve them and will not be modified to try.

What this is not:

- A lesson planning assistant
- A rubric generator

- A curriculum alignment tool
 - A feedback system
 - A system that produces more than one objective
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P2 — Intellectual Foundation

Principle 1: Implementation intentions require specificity to fire Gollwitzer (1999)

established that behavioral intentions convert to action at significantly higher rates when stated as “when X occurs, I will do Y” rather than as goals. The mechanism: specificity pre-loads the cognitive response pattern before the moment arrives. A session objective that names a concrete, observable success criterion functions as an implementation intention for the teacher. A vague objective does not.

Product mechanism: The success criterion field forces behavioral specificity — it must be an observable action with a threshold count (“can do X three times without prompting”), not a cognitive state (“understands X”).

What breaks if compromised: If success criterion is permitted to be abstract (“demonstrates understanding”), the objective cannot be falsified and reverts to intent. The entire governing function of the tool collapses.

Principle 2: Constraint form prevents scope expansion An objective stated as what to maximize implicitly names what to sacrifice. Without the constraint field (must not break), teachers expand scope mid-session — trading depth for coverage. The constraint field makes the trade explicit and locked before the session begins.

Product mechanism: The must-not-break field is required, not optional. Its absence is a validation error, not a missing feature.

What breaks if compromised: Without it, the crystallized objective is a goal, not a constraint system. Scope creep is unblocked.

P3 — The User

The moment, precisely: A teacher with 5–10 minutes before a session begins. They know the content they intend to cover. They have not articulated a success criterion. They are not in a reflective state — they are in a preparation state, time-pressured, cognitively loaded with logistics.

Cognitive/emotional state at moment of use: Low-reflection. High-action orientation.

They will not read instructions. They will not iterate on prompts. They will type one thing and expect one useful thing back.

What they need in the next 60 seconds: One sentence they can say to open the session that is more precise than their intent, without requiring them to understand why it is more precise.

What they do not need:

- A list of options to choose from
- A request for more information before output is produced
- Praise for their stated intent
- An explanation of what a falsifiable objective is
- More than one output

Success from their perspective: They read the crystallized objective, recognize it as a sharper version of what they meant, and use it to open the session. They do not edit it. They do not have to think about it.

P4 — Core Experience

Step 1 — Intent capture *Behavioral job:* Lower the cost of input to zero. *Mechanism:* Single textarea. No format requirements. No character limit. No examples shown that might anchor the teacher's framing. *Constraint:* Must accept any natural language form — single word, run-on paragraph, pedagogical jargon, colloquial phrasing. *Failure mode:* If the field imposes structure (dropdowns, required fields beyond the intent itself), teachers will fill it with what the field expects rather than their actual intent.

Step 2 — Crystallization *Behavioral job:* Convert intent to constraint-form objective. One API call. No intermediate state shown to the teacher. *Mechanism:* System prompt enforces three-field structure. Output validated for behavioral specificity of success criterion before returned. *Constraint:* If the input cannot produce a falsifiable success criterion, the system returns one clarifying question — never a partial objective, never a placeholder. *Failure mode:* Returning an objective with an abstract success criterion ("when they demonstrate understanding") produces the appearance of crystallization while leaving the intent unresolved.

Step 3 — Output delivery *Behavioral job:* Make the objective usable in under 10 seconds. *Mechanism:* Three labeled fields rendered as a single readable block: the teacher can read it top-to-bottom as a sentence. *Constraint:* No buttons to "refine," "regenerate," or "explore"

alternatives." One output. Edit state available but not foregrounded. *Emotional register at exit*: Clarity. Not inspiration. Not enthusiasm. Clarity.

Step 4 — Signal capture *Behavioral job*: Detect whether crystallization worked.

Mechanism: Log whether teacher edited the output and to what degree before session start.

Constraint: Edit tracking must not require any teacher action — passive only.

P5 — Output System

Structure:

OBJECTIVE: [maximize field — what to optimize for, ≤12 words]

CONSTRAINT: [must not break — what cannot be sacrificed, ≤12 words]

YOU PASS WHEN: [behavioral criterion — observable action + threshold, ≤15 words]

Derivation logic per field:

- *Maximize*: Extracted from the highest-order goal implicit in the intent statement
- *Must not break*: Derived from the most common trade-off teachers make when pursuing that goal (coverage vs. depth, fluency vs. accuracy)
- *Success criterion*: Must name a specific observable behavior and a repetition threshold without prompting; if the intent contains no behavioral signal, system generates one based on the pattern type and asks one yes/no confirmation before output is finalized

Minimum acceptable output: All three fields populated, success criterion behaviorally specific, no field exceeds its word limit.

Output failure definition: A success criterion that names a cognitive state ("understands," "grasps," "demonstrates comprehension") rather than an observable action. This is the only failure mode that invalidates the output. The system must not return it.

P6 — Measurement System

Primary metric: Edit rate — percentage of crystallized outputs used without modification.

Secondary metric: 72-hour return rate — does the teacher use the tool again within three sessions?

Measurement threshold: Edit rate >40% signals the prompt is wrong. Rewrite the prompt engineering before next iteration. Do not modify the UI, the UX flow, or the output format.

The prompt is the product.

Explicitly not permitted uses of measurement:

- Measuring teacher satisfaction (satisfaction $\neq$ pedagogical effectiveness)
 - Measuring session outcomes (outside scope of this MVP — no learner data)
 - Surfacing edit rate to the teacher (creates self-consciousness that alters behavior)
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P8 — Strategic Constraints & Non-Negotiables

- One output per request. Never two. Never “here are two versions.”
 - The success criterion must be behavioral. If it cannot be made behavioral, return one clarifying question, not a placeholder.
 - No praise for teacher input. No “great idea!” No filler before the output.
 - The must-not-break field is required. Its absence returns a validation error.
 - Edit tracking is passive. No teacher action required.
 - No regeneration button in WEDGE phase. One shot. This is the signal that reveals whether the prompt works.
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P9 — Metrics & Success Definition

Primary hypothesis: A constraint-form objective, system-crystallized from natural-language intent, will be used without modification by >60% of teachers in the first session of use.

Falsification threshold: <40% use-as-is rate after 50 sessions = prompt is wrong. Not UX. Not concept. Prompt.

Phase exit criterion (WEDGE → NEXT): >60% use-as-is rate sustained over three weeks AND >50% 72-hour return rate.

What the team learns: Whether the system can derive a falsifiable success criterion from unstructured teacher intent reliably enough to eliminate the need for teacher reflection at the moment of session preparation.